Electromagnetics I Midterm Exam.

(11/01/2023) Class:_____ Student ID.:_____ Name:_____

A. Selections (50%)

1.(2) Given vectors $\mathbf{A} = \mathbf{a}_x + \mathbf{a}_y 2 - \mathbf{a}_z 3$ and $\mathbf{B} = \mathbf{a}_x 5 + \mathbf{a}_z 2$. What is the component of $\mathbf{A}$ in the direction of $\mathbf{B}$?

[1.]$\frac{1}{\sqrt{29}}$ [2.]$\frac{-1}{\sqrt{29}}$ [3.]$\frac{6}{\sqrt{29}}$ [4.]$\frac{-1}{\sqrt{14}}$ [5.]$\frac{1}{\sqrt{14}}$ [6.] None of the above.

Sol: $\mathbf{A} \cdot \mathbf{a}_B = \frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{B}|} = \frac{5-6}{\sqrt{25+4}} = \frac{-1}{\sqrt{29}}$

2.(5) Given the point $P: (R, \theta, \phi) = \left(2, \frac{\pi}{6}, \frac{\pi}{3}\right)$, with $\mathbf{R}$ denotes its position vector, please calculate $\mathbf{R} \cdot \mathbf{a}_y = ?$, [1.] $\frac{1}{\sqrt{2}}$; [2.]$\frac{1}{2}$; [3.] 1, [4.] $\sqrt{2}$; [5.]$\frac{\sqrt{3}}{2}$, [6.] None of the above.

$\mathbf{R} \cdot \mathbf{a}_y = R \sin\theta \sin\phi = 2 \sin\left(\frac{\pi}{6}\right) \sin\left(\frac{\pi}{3}\right) = 2 \cdot \frac{1}{2} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$

3.(3) In the cylindrical coordinate system, what is the result of $\frac{\partial \mathbf{a}_\phi}{\partial \phi}$? [1.]$\mathbf{a}_\phi$ [2.]$\mathbf{0}$ [3.]$-\mathbf{a}_r$ [4.]$\mathbf{a}_r$ [5.] $-\mathbf{a}_\phi$ [6.] None of the above.

Sol: $\frac{\partial \mathbf{a}_\phi}{\partial \phi} = \frac{\partial[-\mathbf{a}_x \sin(\phi) + \mathbf{a}_y \cos(\phi)]}{\partial \phi} = -\mathbf{a}_x \cos(\phi) - \mathbf{a}_y \sin(\phi) = -\mathbf{a}_r$

4.(4) Consider a vector function $\mathbf{F} = \mathbf{a}_x(x + c_1 z) + \mathbf{a}_y(c_2 x - 3z) + \mathbf{a}_z(x + c_3 y + c_4 z)$. If $\mathbf{F}$ is irrotational and solenoidal, then $c_4 =$ [1.] 1; [2.] 0; [3.]-3, [4.]-1; [5.] 2, [6.] None of the above.

Sol: refer to Quiz 2

5.(5) Given a scalar function $V = \left(\sin\frac{\pi}{2}x\right)\left(\sin\frac{\pi}{3}y\right)e^{-z}$, find $\nabla \times (\nabla V)$ at the point $P: (x, y, z) = (1,2,3)$.

[1.]$\left(-\mathbf{a}_y \frac{\pi}{6} - \mathbf{a}_z \frac{\sqrt{3}}{2}\right)e^{-3}$; [2.]$\frac{1}{\sqrt{14}}(-\mathbf{a}_x - \mathbf{a}_y 2 - \mathbf{a}_z 3)$; [3.]$-\mathbf{a}_x - \mathbf{a}_y 2 - \mathbf{a}_z 3$, [4.]$\mathbf{a}_x 5 + \mathbf{a}_z 2$; [5.] $\mathbf{0}$, [6.] None of the above.

Sol: The curl of the gradient of a scalar field is always a zero vector.

6.(1) What of the following about a conductor in an electrostatic field is wrong? [1.]$\rho = 0$ at any point on the surface of the conductor; [2.] The conductor is of equipotential; [3.] Outside the conductor, $\mathbf{E}$ is normal to the conductor surface; [4.] $\mathbf{E} = \mathbf{0}$ inside the conductor; [5.] $\rho = 0$ inside the conductor.

7.(3) Let C be a counter-clockwise contour along a circle on the x-y plane with radius 2 and centered at the origin, please calculate $\oint_C \boldsymbol{a}_\phi \cdot d\boldsymbol{\ell}$. [1.]$\mathbf{0}$; [2.] $2\pi \mathbf{a}_\phi$ [3.] 4π; [4.] $4\pi \boldsymbol{a}_r$; [5.] None of the above.

Sol: $\oint_C \boldsymbol{a}_\phi \cdot d\boldsymbol{\ell} = \int_0^{2\pi} \mathbf{a}_\phi \cdot (\mathbf{a}_\phi 2 d\phi) = 4\pi$

please calculate $\oint_C \mathbf{a}_\phi \cdot (\mathbf{a}_\phi 2 d\phi) = 4\pi$

8.(4) Given that in free space a point charge q is placed at the position $P_0: (x, y, z) = (1,0,-3)$, what is the resultant $\boldsymbol{E}$ at point $P_1: (x, y, z) = (1,0,0)$, [1.] $-\boldsymbol{a}_z \frac{q}{36\pi\varepsilon_0}$; [2.] $-\boldsymbol{a}_z \frac{q}{12\pi\varepsilon_0}$; [3.] $\boldsymbol{a}_R \frac{q}{4\pi\varepsilon_0 R^2}$; [4.] $\boldsymbol{a}_z \frac{q}{36\pi\varepsilon_0}$; [5.] None of the above.

9.(2) A surface charges with density $\rho = \rho_s$ is uniformly distributed in air on a spherical surface of radius 3 centered at the origin. Please calculate the electric potential V at position $P:(R,\theta,\phi) = \left(4, \frac{\pi}{4}, 0\right)$. [1.] $\frac{\rho_s}{9\varepsilon_o}$; [2.] $\frac{9\rho_s}{4\varepsilon_o}$; [3.] $\frac{15\rho_s}{27\varepsilon_o}$;

[4.] $\frac{15\rho_s}{27\varepsilon_o}$; [5.] 0; [6.]None of the above.

Sol: Due to the symmetry condition, we know that $\mathbf{E}$ only has $\mathbf{a}_R E_R$ component. So, by using Gauss's Theorem we can find $E_R = \frac{9\rho_s}{\varepsilon_o R^2}$. $\Rightarrow V(R) = -\int_\infty^R \mathbf{a}_R \frac{9\rho_s}{\varepsilon_o r^2} \cdot \mathbf{a}_R dr = \frac{9\rho_s}{\varepsilon_o R}$. Thus, at P, $V = \frac{9\rho_s}{4\varepsilon_o}$

Constructing a spherical Gaussian surface with radius $R > 3$,

$\mathbf{E} = \mathbf{a}_r E_R$, $d\mathbf{S} = \mathbf{a}_r R^2 \sin\theta d\theta d\phi$, $\mathbf{E} \cdot d\mathbf{S} = E_R R^2 \sin\theta d\theta d\phi$

$\oint_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q}{\varepsilon_0} \Rightarrow \int_0^{2\pi}\int_0^{\pi} E_R R^2 \sin\theta d\theta d\phi = \frac{4\pi(3)^2 \rho_s}{\varepsilon_0}$

$\Rightarrow 4\pi R^2 E_R = \frac{4\pi(3)^2 \rho_s}{\varepsilon_0} \Rightarrow E_R = \frac{9\rho_s}{R^2 \varepsilon_0}$

[4.] $\frac{15\rho_s}{27\varepsilon_0}$; [5.] 0; [6.]None of the above.

Sol: Due to the symmetry condition, we know that $\mathbf{E}$ only has $\mathbf{a}_R E_R$ component. So, by using Gauss's Theorem we can find $E_R = \frac{9\rho_s}{\varepsilon_0 R^2}$. $\Rightarrow V(R) = -\int_\infty^R \mathbf{a}_R \frac{9\rho_s}{\varepsilon_0 r^2} \cdot \mathbf{a}_R dr = \frac{9\rho_s}{\varepsilon_0 R}$. Thus, at P, $V = \frac{9\rho_s}{4\varepsilon_0}$

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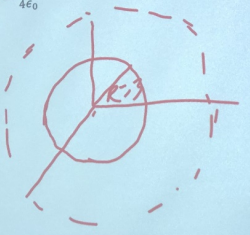
$\oint_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q}{\varepsilon_0} \Rightarrow \int_0^{2\pi}\int_0^{\pi} E_R R^2 \sin\theta d\theta d\phi = \frac{4\pi(3)^2 \rho_s}{\varepsilon_0}$

$\Rightarrow 4\pi R^2 E_R = \frac{4\pi(3)^2 \rho_s}{\varepsilon_0} \Rightarrow E_R = \frac{9\rho_s}{R^2 \varepsilon_0}$



At $P:(R,\theta,\phi) = \left(4, \pi/4, 0\right)$, $\mathbf{E} = \mathbf{a}_R \frac{9\rho_s}{16\varepsilon_0}$

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10.(1) Same as the above problem, please calculate the $\boldsymbol{E}$ value at position $P:(R,\theta,\varphi) = \left(1, \frac{3\pi}{4}, \frac{\pi}{2}\right)$ [1.] **0**,

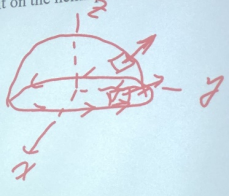
[2.] $\boldsymbol{a}_R \frac{9\rho_s}{\varepsilon_0}$; [3.] $\boldsymbol{a}_R \frac{\rho_s}{27\varepsilon_0}$; [4.] $\boldsymbol{a}_R \frac{\rho_s}{9\varepsilon_0}$, [5.] $-\boldsymbol{a}_R \frac{9\rho_s}{\varepsilon_0}$; [6.] None of the above.

Constructing a spherical Gaussian surface with radius $R < 3$,

Sol: $\oint_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q}{\varepsilon_0} = 0$, $\Rightarrow$ At $P:(R,\theta,\phi) = \left(1, 3\pi/4, \pi/2\right)$, $\mathbf{E} = \mathbf{0}$

B. Calculations (50%)

1. Verify Stokes's Theorem for the vector field $\mathbf{A} = \mathbf{a}_R \cos\theta + \mathbf{a}_\phi \sin\theta$ by evaluating it on the hemisphere of unit radius in the spherical coordinate system. (10%)



Solution:

$$\mathbf{A} = \mathbf{a}_R \cos\theta + \mathbf{a}_\phi \sin\theta = \mathbf{a}_R A_R + \mathbf{a}_\theta A_\theta + \mathbf{a}_\phi A_\phi.$$

Hence, $A_R = \cos\theta, A_\theta = 0, A_\phi = \sin\theta$.

$$\nabla \times \mathbf{A} = \frac{1}{R^2 \sin\theta} \begin{vmatrix} \mathbf{a}_R & R\mathbf{a}_\theta & R\sin\theta\, \mathbf{a}_\phi \\ \frac{\partial}{\partial R} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ \cos\theta & 0 & R\sin^2\theta \end{vmatrix}$$

$$= \mathbf{a}_R \frac{1}{R^2 \sin\theta} \frac{\partial}{\partial\theta}(R\sin^2\theta) - \frac{1}{R^2\sin\theta}(R\mathbf{a}_\theta)\frac{\partial}{\partial R}(R\sin^2\theta) - \frac{1}{R^2\sin\theta}(R\sin\theta\,\mathbf{a}_\phi)\frac{\partial}{\partial\theta}(\cos\theta)$$

$$= \mathbf{a}_R \frac{2\cos\theta}{R} - \mathbf{a}_\theta \frac{\sin\theta}{R} + \mathbf{a}_\phi \frac{\sin\theta}{R}.$$

$$\int_{\phi=0}^{2\pi}\int_{\theta=0}^{\pi/2}(\nabla\times\mathbf{A})\cdot ds$$

$$=\int_{\phi=0}^{2\pi}\int_{\theta=0}^{\pi/2}\left(\frac{\mathbf{a}_R 2\cos\theta}{R}-\mathbf{a}_\theta\frac{\sin\theta}{R}+\mathbf{a}_\phi\frac{\sin\theta}{R}\right)\cdot\mathbf{a}_R R^2\sin\theta d\theta d\phi\Big|_{R=1}$$

$$=4\pi R\frac{\sin^2\theta}{2}\Big|_0^{\pi/2}\Big|_{R=1}=2\pi$$

The contour C is the (counter-clock-wise) circle in the x-y plane bounding the hemispherical surface.

$$\oint_C \mathbf{A}\cdot d\mathbf{l}=\int_{\phi=0}^{2\pi}(\mathbf{a}_R\cos\theta+\mathbf{a}_\phi\sin\theta)\cdot\mathbf{a}_\phi R\sin\theta\, d\phi\Big|_{\substack{\theta=\pi/2\\R=1}}=R\sin^2\theta\int_0^{2\pi}d\phi\Big|_{\substack{\theta=\pi/2\\R=1}}=2\pi$$

2. A finite line charge of length L carrying uniform line charge density ρ_ℓ is coincident with the x-axis. (a) Determine V in the plane bisecting the line charge. (b) Determine $\mathbf{E}$ from ρ_ℓ directly by applying Coulomb's law. (10%)

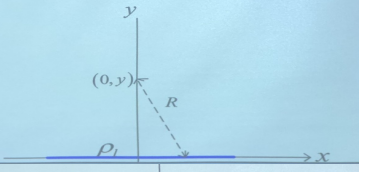
Sol: refer to Problem 3-16

P.3-16

⟨Sol.⟩

(a.)

$\mathbf{R}=-\mathbf{a}_x x+\mathbf{a}_y y, \Rightarrow R=(x^2+y^2)^{\frac{1}{2}},\ dl=dx,\ -\frac{L}{2}\le x\le\frac{L}{2}$

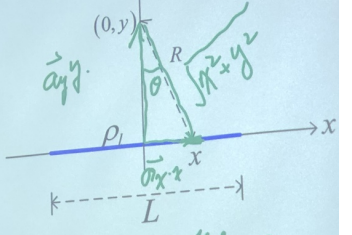


P.3-16

⟨Sol.⟩

(a.)

$\mathbf{R}=-\mathbf{a}_x x+\mathbf{a}_y y, \Rightarrow R=(x^2+y^2)^{\frac{1}{2}},\ dl=dx,\ -\frac{L}{2}\le x\le\frac{L}{2}$



$$V=\boxed{\frac{1}{4\pi\varepsilon_0}\int_L\frac{\rho_l}{R}dl}=\frac{\rho_l}{4\pi\varepsilon_0}\int_{-\frac{L}{2}}^{\frac{L}{2}}\frac{1}{\sqrt{x^2+y^2}}dx$$

$\frac{\cos\theta}{y}$

$$=\frac{\rho_l}{4\pi\varepsilon_0}\int_{-\tan^{-1}\frac{L}{2y}}^{\tan^{-1}\frac{L}{2y}}\sec\theta d\theta$$

$\Rightarrow \frac{\sec\theta(\sec^2\theta+\tan^2\theta)}{\sec(\theta+\tan\theta)}d\theta$

$\tan\theta=\frac{x}{y}\Rightarrow x=y\tan\theta$

$dx=y\sec^2\theta\, d\theta$

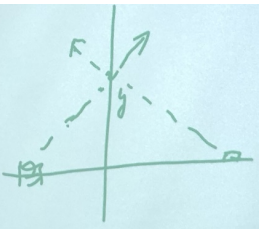
$$=\frac{\rho_l}{2\pi\varepsilon_0}\ln\left(\sqrt{1+\left(\frac{L}{2y}\right)^2}+\frac{L}{2y}\right),\ (\mathrm{V})$$

$2\pi\varepsilon_0$

(b.)

$$\mathbf{E}=\frac{1}{4\pi\varepsilon_0}\int_L \frac{\rho_l}{R^3}dl=\frac{\rho_l}{4\pi\varepsilon_0}\left[\underbrace{\int_{-\frac{L}{2}}^{\frac{L}{2}}\frac{-\mathbf{a}_x x}{\left(x^2+y^2\right)^{\frac{3}{2}}}dx}_{\text{odd fn.}=0}+\int_{-\frac{L}{2}}^{\frac{L}{2}}\frac{\mathbf{a}_y y}{\left(x^2+y^2\right)^{\frac{3}{2}}}dx\right]$$

dQ



$$=\frac{\mathbf{a}_y\rho_l}{2\pi\varepsilon_0 y}\sin\left(\tan^{-1}\frac{L}{2y}\right)$$

$$=\frac{\mathbf{a}_y\rho_l}{2\pi\varepsilon_0 y}\underbrace{\left[\tan\left(\tan^{-1}\frac{L}{2y}\right)\right]}_{\frac{L}{2y}}\underbrace{\cos\left(\tan^{-1}\frac{L}{2y}\right)}_{\frac{1}{\sec\left(\tan^{-1}\frac{L}{2y}\right)}}$$

$$=\frac{\mathbf{a}_y\rho_l}{2\pi\varepsilon_0 y}\frac{L}{2y}\frac{1}{\sqrt{1+\left(\frac{L}{2y}\right)^2}}$$

$$=\mathbf{a}_y\frac{\rho_l L}{4\pi\varepsilon_0 y}\frac{1}{\sqrt{y^2+\left(\frac{L}{2}\right)^2}},\quad (\text{V/m})$$

Draw | Read aloud | Ask Copilot | 10 of 12

ρ_{sb}, respectively.(a)Determine $\mathbf{E}$ everywhere.(b) What must be the relation between b and a in order that $\mathbf{E}$ vanishes for $r>b$?(15%)

⟨Sol.⟩

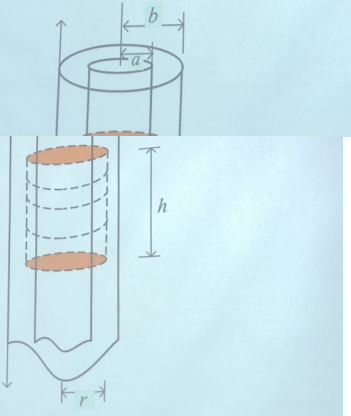
(a.)

Using (r,ϕ,z) coordinate system

$\because$ Infinitely extended along z-axis

$\Rightarrow$ cylindrical symmetric, $\Rightarrow \mathbf{E}=\mathbf{a}_r E_r$

Constructing a cylindrical Gaussian surface of radius r and height h,



$$\Rightarrow \oint_S \mathbf{E}\cdot d\mathbf{s}=\frac{Q}{\varepsilon_0}$$

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- top and bottom surface, $\Rightarrow d\mathbf{s}=\pm\mathbf{a}_z ds, \Rightarrow \mathbf{E}\cdot d\mathbf{s}=0$
- $\oint_S \mathbf{E}\cdot d\mathbf{s}=\int_{\text{side surface}}\mathbf{E}\cdot d\mathbf{s}=\frac{Q}{\varepsilon_0},\ d\mathbf{s}=\mathbf{a}_r r d\phi dz$

(i) $r<a, \Rightarrow Q=0, \Rightarrow \mathbf{E}=0$ ②

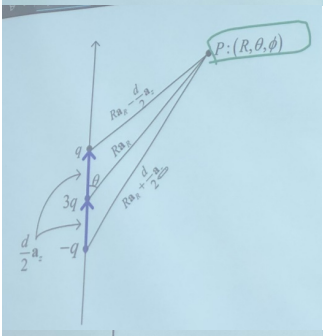
(ii) $a<r<b, \Rightarrow \int_{\text{side surface}}\mathbf{E}\cdot d\mathbf{s}=\int_0^h\int_0^{2\pi}E_r r d\phi dz$

$$=2\pi rhE_r=\frac{2\pi ah\rho_{sa}}{\varepsilon_0}$$

$$\Rightarrow E_r=\frac{a\rho_{sa}}{r\varepsilon_0}$$

(iii) $r>b, \Rightarrow \int_{\text{side surface}}\mathbf{E}\cdot d\mathbf{s}=2\pi rhE_r=\frac{2\pi(a\rho_{sa}+b\rho_{sb})h}{\varepsilon_0},$

4. Three charges $(+q, 3q, \text{and } -q)$ are arranged along the z-axis at $z = \frac{d}{2}$, $z = 0$, and $z = -\frac{d}{2}$, respectively. At the point $P: (R, \theta, \phi)$, please calculate (a) $V = ?$, and (b) $\mathbf{E} = ?$ (15%)



$$Q1 = +q,\ \mathbf{R}_1 = R\mathbf{a}_R - \frac{d}{2}\mathbf{a}_z \Rightarrow \frac{1}{R_1} = \left(R^2 - Rd\cos\theta + \left(\frac{d}{2}\right)^2\right)^{-\frac{1}{2}} \approx R^{-1}\left(1 + \frac{d}{2R}\cos\theta\right)$$

$$Q2 = 3q,\ \mathbf{R}_2 = R\mathbf{a}_R \Rightarrow \frac{1}{R_2} = R^{-1},$$

$$Q3 = -q,\ \mathbf{R}_3 = R\mathbf{a}_R + \frac{d}{2}\mathbf{a}_z \Rightarrow \frac{1}{R_3} = \left(R^2 + Rd\cos\theta + \left(\frac{d}{2}\right)^2\right)^{-\frac{1}{2}} \approx R^{-1}\left(1 - \frac{d}{2R}\cos\theta\right)$$

Taylor's series

$Q2 = 3q,\ \mathbf{R}_2 = R\mathbf{a}_R \Rightarrow \frac{1}{R_2} = R^{-1},$

$Q3 = -q,\ \mathbf{R}_3 = R\mathbf{a}_R + \frac{d}{2}\mathbf{a}_z \Rightarrow \frac{1}{R_3} = \left(R^2 + Rd\cos\theta + \left(\frac{d}{2}\right)^2\right)^{-\frac{1}{2}} \approx R^{-1}\left(1 - \frac{d}{2R}\cos\theta\right)$

(a.)

$$V = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{R_1} + \frac{3}{R_2} + \frac{-1}{R_3}\right]$$

$$= \frac{q}{4\pi\varepsilon_0 R}\left[\left(1 + \frac{d}{2R}\cos\theta\right) + 3 - \left(1 - \frac{d}{2R}\cos\theta\right)\right] = \frac{q}{4\pi\varepsilon_0 R}\left(3 + \frac{d}{R}\cos\theta\right),\ \text{(V)}$$

(b.)

$$\mathbf{E} = -\nabla V = -\mathbf{a}_R\frac{\partial V}{\partial R} - \mathbf{a}_\theta\frac{\partial V}{R\cdot\partial\theta} = \frac{q}{4\pi\varepsilon_0 R^2}\left[\left(3 + \frac{2d}{R}\cos\theta\right)\mathbf{a}_R + \frac{d}{R}\sin\theta\,\mathbf{a}_\theta\right],\ \text{(V/m)}$$